

Future Roadmap

Our current prototype successfully demonstrates EEG-based wheelchair navigation, obstacle detection, and verbal communication. In future versions, we plan to incorporate several advanced features to improve usability, safety, and real-time monitoring.

Phase 1 – Smart Monitoring

GPS Integration

A GPS module will be integrated to continuously track the real-time location of the wheelchair. This feature will assist caregivers and family members in locating the user during emergencies.

SOS Emergency Alert

An SOS alert mechanism will be implemented to send emergency notifications along with the user's live GPS location to registered caregivers or family members whenever the emergency button is pressed or an abnormal condition is detected.

Battery Health Monitoring

We plan to integrate the **INA226 High-Precision Current and Voltage Sensor** to continuously monitor:

- Battery percentage
- Battery voltage
- Current consumption
- Remaining backup time
- Power usage statistics

This will help users avoid unexpected battery failures during operation.

Phase 2 – IoT Dashboard

A dedicated web portal will be developed to display live wheelchair information.

The dashboard will provide:

- Live GPS location
- Battery percentage
- Battery voltage
- Current consumption
- Wheelchair status

- Movement direction
- Obstacle detection status
- System health monitoring

Authorized caregivers will be able to remotely monitor the wheelchair in real time.

Phase 3 – AI Enhancements

Future AI capabilities include:

- Voice command support
- Adaptive navigation using AI
- Personalized user profiles
- Predictive maintenance
- Automatic route optimization
- Intelligent obstacle avoidance

Phase 4 – Medical Integration

Future medical features include:

- Heart rate monitoring
- SpO₂ monitoring
- Fall detection
- Emergency healthcare notifications
- Hospital integration
- Electronic health record synchronization

Long-Term Vision

Our vision is to transform the prototype into a comprehensive intelligent assistive mobility platform that combines Brain-Computer Interface (BCI), Edge AI, IoT, and healthcare technologies to enhance the independence and quality of life of individuals with severe motor disabilities.